

GxP / CGTO / ATO / eSignature

Reuse Components Reference Guide

SAP S/4HANA + BTP | Life Science | 2026-06-24

Purpose of This Document

This document identifies and describes all reuse components referenced across the GxP Compliance, CGTO/ATO Integration, eSignature, and Reauthentication project for SAP S/4HANA Purchase Order. Sources: CGTO PO Alignment meeting (2026-06-12), ATO to S4 PuC Integration Demo (2026-06-17), and Reauthentication Framework session (2026-06-18).

All Reuse Components at a Glance

#	Component	Type	System	Status
1	BTP eSignature SaaS	SaaS Service	BTP	Available — integration planned 2702
2	Reauthentication Framework	ABAP Platform Layer	S4	Available — LOB implements callback
3	Integration Suite iFlow (CPI)	Middleware	BTP Integ. Suite	Built for private cloud — porting to public
4	Event Mesh	Async Event Bus	BTP	Active
5	Change Document Framework	S4 Standard Framework	S4	Verified for PO 2608 ✓ — open issue on ATO fields
6	PurchaseOrder OData API	S4 Standard API	S4	Extension enabled by PO team on request
7	Industry Object Extension	Architecture Pattern	S4	Approved — first public cloud use by ATO
8	Scope Item 84P	SAP Content Delivery	S4 + BTP	Planned — LOBs contribute field configs

■ Open Issue: Change Document display for ATO extension fields is NOT visible in ME22N/ME23N change log screen. ATO extension table /ATO/S4_PO_HEADER_EXT is written to CDPOS but does not surface in the standard PO change log UI. A potential BADI 'Change Document Processing' was identified by Jiss (PO team) — needs investigation. Joint call required: ATO team + PO standard team + Change Document team.

Reuse Component Details

1 BTP eSignature SaaS Service

SaaS Service | BTP | Available — integration planned 2702

A shared, reusable SaaS service already available on BTP. It does NOT need to be built per LOB — it exists as a single instance that any S4 application can integrate with via a Communication Arrangement. It captures re-authentication credentials, creates a tamper-proof signature record, and links it to the S4 Change Document number (CHANGENR).

Owner	Yadesh Gupta / Prabir Kumar Mallick (framework) + Swarnava Chatterjee (eSignature product)
What it stores	Signature ID, User ID, Timestamp, Reason Code, Credential hash — tamper-proof
Auth methods	SAP password, PIN, biometric, external IdP — configurable per object
Integration	Communication Arrangement in S4 — same pattern as SAP_COM_0267
Consumers	ME21N (WS1), Sales Order, Process Order, Maintenance Order — all share the same service
Key rule	Must re-enter credentials — existing session token is NOT sufficient for GxP compliance

"The eSignature reusable service is already available as a SaaS on BTP. It does NOT need to be built — it needs to be integrated into LOB applications."

2 Reauthentication Reuse Framework (S4 Platform Layer)

ABAP Platform Framework | S4 | Available — LOB implements one callback class

A platform-level reusable ABAP framework in S4 that any LOB application registers with to enforce GxP reauthentication. The design goal is one common UX and one common platform layer — every S4 application that needs GxP reauthentication integrates here instead of building its own signature flow. The LOB only implements ONE callback class (Relevancy Check). Everything else — calling BTP, capturing signature, linking to Change Doc — is handled by the framework.

S-table config	VC_SRA_SOT — registers: object name, callback class, auth class, node types, field list
APIs exposed	REQUEST_REAUTH (OData variant), REQUEST_REAUTH (library), VALIDATE_RESULT, CHECK_ESIG_ACTIVE
Utility method	Framework provides utility to read regulated field list from S-table — LOB does not query directly
Display reauth	Separate dedicated class for display-mode reauthentication (no edit event)
LOB implements	Only ONE callback class: Relevancy Check → returns ABAP_TRUE/FALSE
Warning	If Relevancy Check returns ABAP_FALSE due to a bug → reauthentication silently skipped
Confirmed users	Purchase Order (WS1), Process Order (Asset Mgmt — Niranjana/Ajith), Sales Order, Maintenance Order

"We want to have a reuse framework at least from our platform layer so that each and every application which is there in S4 side can integrate to this GxP compliance thing and can have a common user experience for everybody. — Yadesh Gupta"

3 Integration Suite iFlow (CPI — Middleware)

Middleware | BTP Integration Suite | Built for private cloud — being ported to public cloud

The iFlow is the middleware layer between ATO SaaS (BTP) and S4. It is required because ATO SaaS and S4 have DIFFERENT event schemas — the iFlow bridges the schema gap in both directions. It is not a replication layer; it performs schema mapping and event routing only.

Why needed	ATO BTP schema and S4 event schema are different — iFlow maps between them in both directions
Directions	BTP→S4: maps ATO event payload → S4 PurchaseOrder API format. S4→BTP: maps PO change event → ATO schema
Not replication	Confirmed by Loring Wu: 'It is not a replication, it is just a linkage between the documents'
Subscriber acct	BTP subscriber account publishes/receives from Event Mesh via iFlow
Scope	Handles PO create, PO update, PO cancel events. Also Sales Order, Process Order, Production Order
Public cloud gap	Private cloud used legacy BAPI-based iFlows — being redesigned for public cloud OData API pattern

"We have the middleware CPI. CPO will convert the payload according to S4 purchase order API and call the API to create the purchase order. — Loring Wu"

4 BTP Event Mesh

Async Event Bus | BTP | Active

SAP BTP Event Mesh is the shared async messaging infrastructure used for all bidirectional event exchange between ATO SaaS (BTP) and S4. It decouples the two systems so they operate independently — events are published and consumed asynchronously.

Events from S4	PurchaseOrder.Created, PurchaseOrder.Changed — includes standard fields + ATO extension fields
Events from ATO	PO Create request, PO Update request (collection qty, patient ID, protocol ID), PO Cancellation
Pattern	Publish-subscribe — ATO SaaS subscribes to S4 PO events via subscriber account
Outbound events	S4 publishes outbound event on every PO change relevant to ATO — even when user edits in ME21N
Scope	Also used for Sales Order, Process Order, Production Order, Stock events

Reuse Component Details (continued)

5 Change Document Framework

S4 Standard Framework | S4 | Verified for PO 2608 ✓ — open issue on ATO extension fields

The standard SAP Change Document framework (CDHDR/CDPOS) is the audit trail backbone for GxP compliance. It records who changed what, when, with old and new values. It is the PREREQUISITE for eSignature — eSignature links its signature record to the Change Document number (CHANGENR). Without Change Document, eSignature has no anchor point.

Tables	CDHDR: change doc header (object class, PO number, user, date/time). CDPOS: one row per changed field with VALUE_OLD/VALUE_NEW
ATO registration	ATO extension table /ATO/S4_PO_HEADER_EXT must be explicitly registered in the CD object to appear in CDPOS
eSign dependency	Signature ID from BTP is linked to CHANGENR — CD must fire before eSignature can attach
Verified 2608	Change Document verified working for all standard PO fields in release 2608 ✓
Open issue	ATO extension field changes appear in CDPOS but are NOT visible in ME22N/ME23N change log UI
Possible BAdI	Jiss (PO team) identified BAdI 'Change Document Processing' — may surface extension changes. Not confirmed.
Action needed	Joint call: ATO team (Allen Yuan/Loring Wu) + PO standard team (Jiss) + Change Document team

"For the change log, we needed to extend the standard change document object to include our ATO table, then the change can be recorded. But since the object is a separate ATO table, no description is displayed in ME22N directly. — Allen Yuan (ATO team)"

6 PurchaseOrder OData API Extension

S4 Standard API — Extended | S4 | Extension enabled by PO team on ATO request

The standard S4 PurchaseOrder OData V4 API extended with ATO industry fields. ATO does NOT have a separate standalone API — it extends the standard one so that standard PO fields and ATO extension fields are created/updated in a single API call. The extension was NOT enabled by default and required the PO standard team to enable it.

Extension type	Industry Object Extension node associated to PurchaseOrder entity — TherapyType, ClinicalStudy, ProtocolRef, PatientRef
Why not separate	Extension node data is dependent on PO — cannot exist without the standard document. Separate API would break this dependency.
Who enabled it	Nils Hartmann / Jiss (PO standard team) — ATO raised limitation during POC, PO team planned and delivered the enablement
ATO usage	ATO calls this single API for PO creation — standard fields + ATO extension fields in one payload
Confirmed by	Loring Wu: 'The extension was not enabled in the purchase order API, so we reached out to your team to enable it'
Governance	Any further API extension changes require Suite Architecture review (Felix Vente — North Star)

"We should not touch or we should not change anything of the standard objects. In case we face any restriction, we need to reach out to your team to enable the extensibility. — Loring Wu"

7 Industry Object Extension Pattern

Architecture Pattern | S4 | Approved — first public cloud use by ATO/CGTO

The SAP-approved extensibility pattern for adding industry-specific fields to standard S4 business objects (EKKO/EKPO) without modifying the standard tables. ATO is the FIRST Life Science industry to use this pattern in public cloud — Discrete and Retail used similar approaches before the code split via a different path.

Why not native	ATO requires THREE levels of extension (header, item, third-level). Native extension supports only two levels.
POC outcome	Loring Wu: 'After POC with native extension and separate BO, conclusion: separate BO because of third-layer extension'
Approved by	Felix Vente (Suite Architecture — North Star governance)
Tables created	/ATO/S4_PO_HEADER_EXT (FK → EKKO.EBELN): TherapyType, ClinicalStudy, ProtocolRef
	/ATO/S4_PO_ITEM_EXT (FK → EKPO.EBELN+EBELP): TherapyType, PatientRef
Key rule	NEVER append fields directly to EKKO/EKPO — always via Industry Object Extension node
Reuse scope	Same pattern applied across PO, Sales Order, Process Order, Production Order — not PO-specific

"We are the first one in public cloud to integrate for creation of purchase order for Life Science. We started working with Nils and Jiss together with the Suite architecture team in public cloud, which is led by Felix Vente. — Sathishkumar Meenakshisundaram"

8 Scope Item 84P — GxP Compliance

SAP Content Delivery | S4 + BTP | Planned — LOBs contribute their field configs

The central SAP scope item for GxP compliance. When activated in S4, it enables the Reauthentication Framework and loads the S-table entries. It also triggers automatic provisioning of reason codes in the BTP tenant. LOB teams (e.g. PO team) ship their specific regulated field lists and S-table configurations inside 84P — all LOBs share the same scope item, not one per LOB.

What it activates	Reauthentication Framework in S4, Communication Arrangement to BTP eSignature, S-table baseline entries
BTP provisioning	Reason codes defined for each business object are auto-loaded in BTP tenant when 84P activates
LOB contribution	Each LOB ships their object registration + node types + regulated field list inside 84P transport
Customer tuning	After activation, customer can fine-tune field list (remove/add fields) and customise reason codes in BTP
All LOBs share	One 84P scope item used by PO, Sales Order, Process Order, Maintenance Order — not duplicated per LOB
Delivery model	S-table content (object + fields) delivered via 84P transport. Reason codes delivered via BTP content

Open Issue — Change Document Display for ATO Extension Fields

What Was Raised

In the ATO to S4 PuC Integration Demo meeting (2026-06-17), Allen Yuan (ATO team) demonstrated a specific challenge: when ATO extension fields (e.g. THERAPY_TYPE) are changed and written to CDHDR/CDPOS, the change log entries are NOT visible in the standard ME22N / ME23N change log screen. This is a reuse challenge — the Change Document Framework works correctly, but the display layer does not surface extension table records alongside standard PO field changes.

Aspect	Detail
What works	ATO extension field changes ARE written to CDPOS with TABNAME=/ATO/S4_PO_HEADER_EXT — data is recorded correctly
What does not	ME22N / ME23N change log screen does not display these CDPOS rows alongside standard PO change rows
Root cause	The standard PO change document object does not include /ATO/S4_PO_HEADER_EXT in its registered table list
Option A	Extend the standard PO change document object to include /ATO/S4_PO_HEADER_EXT — requires standard team involvement
Option B	BAdI 'Change Document Processing' identified by Jiss (PO team) — may surface extension rows. Not yet confirmed.
Sales Order ref	ATO extension for Sales Order handled by RAP/BOPF framework automatically — different path than PO (Web GUI)
Confirmed by	Allen Yuan (ATO), Jiss Nimmi Augustine (PO standard), Nils Hartmann (PO Product Owner)
Action	Joint call required: ATO team (Allen Yuan / Loring Wu) + PO standard team (Jiss) + Change Document team

Why This Matters for GxP

For GxP compliance (21 CFR Part 11), the complete change history of all regulated fields must be visible in a single, unified audit trail view. If the ATO extension field changes (Therapy Type, Clinical Study, Protocol) appear in CDPOS but are invisible in the standard change log screen, auditors cannot see the complete regulated field history in one place. This is a GxP compliance gap that must be resolved before 2702 delivery.

Sources

Meeting	Date	Key Speakers	Relevant Timestamp
CGTO PO Alignment	2026-06-12	Nils Hartmann, Loring Wu, Satish Kumar, Jiss	Throughout
ATO to S4 PuC Integration Demo	2026-06-17	Allen Yuan, Loring Wu, Erwin Sha, Jiss, Nils	32:09 – 44:00
Reauthentication Framework	2026-06-18	Yadesh Gupta, Prabir Kumar Mallick, Swarnava	Throughout